Understanding Transformer Memorization Recall Through Idioms

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Abstract

To produce accurate predictions, language models (LMs) must balance between generalization and memorization. Yet, little is known about the mechanism by which transformer LMs employ their memorization capacity. When does a model decide to output a memorized phrase, and how is this phrase then retrieved from memory? In this work, we offer the first methodological framework for probing and characterizing recall of memorized sequences in transformer LMs. First, we lay out criteria for detecting model inputs that trigger memory recall, and propose *idioms* as inputs that typically fulfill these criteria. Next, we construct a dataset of English idioms and use it to compare model behavior on memorized vs. non-memorized inputs. Specifically, we analyze the internal prediction construction process by interpreting the model's hidden representations as a gradual refinement of the output probability distribution. We find that across different model sizes and architectures, memorized predictions are a two-step process: early layers promote the predicted token to the top of the output distribution, and upper layers increase model confidence. This suggests that memorized information is stored and retrieved in the early layers of the network. Last, we demonstrate the utility of our methodology beyond idioms in memorized factual statements. Overall, our work makes a first step towards understanding memory recall, and provides a methodological basis for future studies of transformer memorization.[1]

1 Introduction

Transformer language models (LMs) memorize instances from their training data (Carlini et al., 2021; Zhang et al., 2021b), and evidence is building that such memorization is an important precondition for their predictive abilities (Lee et al., 2022; Feldman, 2020; Feldman and Zhang, 2020; Raunak et al., 2021; Raunak and Menezes, 2022). Still, it is unknown when models decide to output memorized sequences, and how these sequences are being retrieved internally from memory. Current methods for analyzing memorization (Feldman and Zhang, 2020; Zhang et al., 2021b; Carlini et al., 2022) use definitions that are based on models performance, which changes between models and often also between training runs. Moreover, these methods study memorization behavior in terms of the model's "black-box" behavior rather than deriving a behavioral profile of memory recall itself.

Our first contributions are to provide a definition and construct a dataset that allows probing memorization recall in LMs. We define a set of criteria for identifying memorized sequences that do not depend on model behavior:[2] sequences that have a single plausible completion that is independent of context and can be inferred only given the entire sequence. We show that many idioms (e.g., "play it by ear") fulfill these conditions, allowing us to probe and analyze memorization behavior. Furthermore, we construct a dataset of such English idioms, dubbed IDIOMEM, and release it publicly for the research community.

Next, to analyze memory recall behavior, we compare the construction process of predictions that involve memory recall with those that do not. To this end, given a LM, we create two sets of memorized and non-memorized idioms from IDIOMEM (Fig. 1, A). We then adopt a view of the transformer inference pass as a gradual refinement of the output probability distribution (Geva et al., 2021; Elhage et al., 2021). Concretely, the token repre-

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[1]Our code and data are available at https://github.com/adihaviv/idiomem/.

[2]Literature often purports to "define memorization", resulting in a multitude of technical definitions with subtle differences, although we would expect this concept to be consistent and intuitive. Thus, instead of explicitly defining "memorization", we will define *sufficient criteria* for detecting memorized instances.

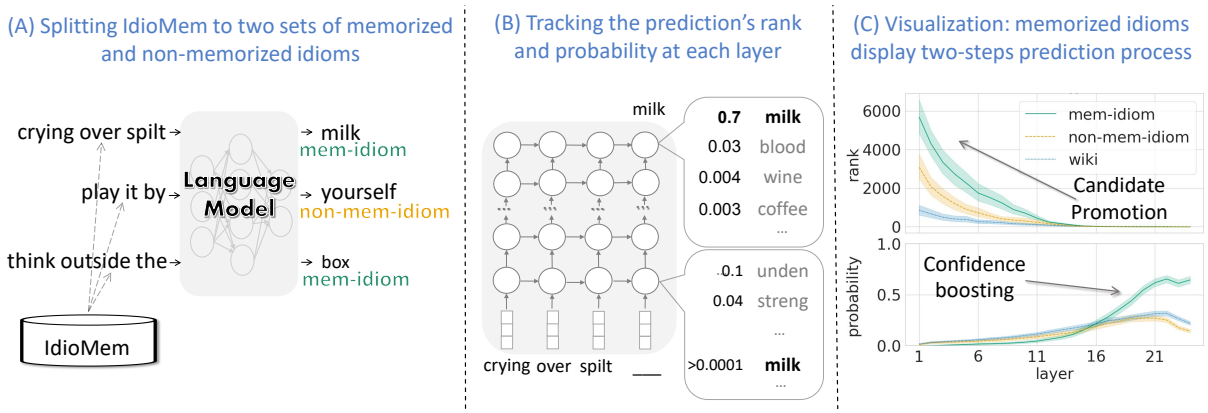

Figure 1: Our methodological framework for probing and analyzing memorized predictions of a given LM: (A) we create two sets of memorized (`mem-idiom`) and non-memorized (`non-mem-idiom`) idioms by probing the LM with instances from IDIOMEM, (B) for each instance, we extract hidden features of the prediction computation – the rank and probability of the predicted token across layers, and (C) we compare the prediction process of memorized idioms versus non-memorized idioms and short sequences from Wikipedia (`wiki`). Memorized predictions exhibit two characteristic phases: candidate promotion and confidence-boosting.

sentation at any layer is interpreted as a "hidden" probability distribution over the output vocabulary (Geva et al., 2022) (Fig. 1, B). This interpretation allows tracking the prediction across layers in the evolving distribution. We find a clear difference in model behavior between memorized and non-memorized predictions (Fig. 1, C). This difference persists across different transformer architectures and sizes: retrieval from memory happens in two distinct phases, corresponding to distinct roles of the transformer parameters and layers: (1) *candidate promotion* of memorized predictions' rank in the hidden distribution in the first layers, and (2) *confidence boosting* where, in the last few layers, the prediction's probability grows substantially faster than before. This is unlike non-memorized predictions, where the two phases are less pronounced and often indistinct. We further confirm these phases of memorized predictions through intervention in the network's FFN sublayers, which have been shown to play an important role in the prediction construction process (Geva et al., 2022; Mickus et al., 2022). Concretely, zeroing out hidden FFN neurons in early layers deteriorate memory recall, while intervention in upper layers does not affect it.

Last, we show our findings extend to types of memory recall beyond idioms by applying our method to factual statements from the LAMA-UHN dataset (Poerner et al., 2020) (e.g. *"The native language of Jean Marais is French"*). For factual statements that were completed correctly by the LM, we observe the same two phases as in memorized idioms, further indicating their connection to memory recall.

To summarize, we construct a novel dataset of idioms, usable for probing LM memorization irrespective of the model architecture or training parameterization. We then design a probing methodology that extracts carefully-devised features of the internal inference procedure in transformers. By applying our methodology and using our new dataset, we discover a profile that characterizes memory recall, across transformer LMs and types of memorized instances. Our released dataset, probing framework, and findings open the door for future work on transformer memorization, to ultimately demystify the internals of neural memory in LMs.

2 Criteria for Detecting Memory Recall

To study memory recall, we require a set of inputs that trigger this process. Prior work on memorization focused on detecting instances whose inclusion in the training data has a specific influence on model behavior, such as increased accuracy on those instances (Feldman and Zhang, 2020; Magar and Schwartz, 2022; Carlini et al., 2022, 2021, 2019). As a result, memorized instances differ across models and training parameterization. Our goal is instead to find a stable dataset of sequences that correctly predicting their completion indicates memorization recall. This will greatly reduce the overhead of studying memorization and facilitate useful comparisons across models and studies.

To build such a dataset, we start by defining a general set of criteria that are predicates on se-

quence features, entirely independent of the LM being probed. Given a textual sequence of n words, we call the first $n-1$ words the *prompt* and the nth word the *target*. We focus on the task of predicting the target given the prompt, i.e., predicting the last word in a sequence given its prefix.[3] Such predictions can be based on either generalization or memorization, and we are interested in isolating memorized cases to study model behavior on them. Particularly, we are looking for *sequences for which success in this task implies memorization recall*.

We argue that the following criteria are *sufficient* for detecting such memorized sequences:

1. **Single target, independent of context:** We require that the target is the only correct continuation, regardless of the textual context where the prompt is placed.[4]

2. **Irreducible prompt:** The target is the single correct completion only if the entire prompt is given exactly. Changing or removing parts from the prompt would make the correct target non-unique.

Claim 2.1. *Assume a sequence fulfills the above criteria. Then, if a LM correctly predicts the target, it is highly likely that this prediction involves memory recall.*

Justification. First, observe that most natural-language prompts have many possible continuations. For example consider the sentence *"to get there fast, you can take this ____"*. Likely continuations include *"route"*, *"highway"*, *"road"*, *"train"*, *"plane"*, *"advice"*, inter alia. Note that there are several divergent interpretations or contexts for the prompt, and for each, language offers many different ways to express similar meaning.

A prediction that is a product of generalization — i.e., it is derived from context and knowledge of language — *always has plausible alternatives*, depending on the context and stylistic choice of words. Hence, the relationship between the entire prompt and the target, where the target is the single correct continuation, is something that needs to be memorized rather than derived via generalization. A LM that predicts *the* single correct continuation either memorized this relationship, or used "cues" from the prompt that happen to provide indication towards the correct continuation. To illustrate the latter, consider the sequence *"it's raining cats and ____"* which has a single correct continuation, *"dogs"*, but a LM might predict it without observing this sequence during training, due to the semantic proximity of *"cats"* and *"dogs"*. Our second criterion excludes such cases by requiring that the correct continuation is only likely given the entire sequence.

Therefore, a LM that correctly completes a sequence that fulfills both criteria, is likely to have recalled it from memory. □

In the next section, we argue that idioms are a special case of such sequences, and are thus useful for studying memorization (§3).

3 The Utility of Idioms for Studying Memorization

An idiom is a group of words with a meaning that is not deducible from the meanings of its individual words. For example, consider the phrase *"play it by ear"* — there is a disconnect between its nonsensical literal meaning (to play something by a human-body organ called 'ear') and its intended idiomatic meaning (to improvise).

A key observation is that *idioms often satisfy our criteria (§2), and therefore can probe memorization*. First, by definition, idioms are expected to be non-compositional (Dankers et al., 2022). They are special "hard-coded" phrases that carry a specific meaning. As a result, their prompts each have a single correct continuation, regardless of their context (criterion 1). For example, consider the prompt *"crying over spilt ____"* — a generalizing prediction would predict that this slot may be filled by any spillable item, like wine, water or juice, while a memorized prediction will retrieve only milk in this context. Notably, while this is an empirical characterization of many idioms, there might be exceptions, e.g., contexts that are adversarially chosen to change the completion. Second, many idioms are "irreducible", for example, the sub-sequences *"crying over"* or *"over spilt"* by themselves have but a scant connection to the word *"milk"*.

Still, not all idioms fulfill the criteria. For example, even when the idiom is far from literal, its constituents sometimes strongly indicate the correct continuation, such as with the case of *"it's raining cats and ____"* (as explained in §2). To construct

[3] In cases where tokenization divides the target to subtokens, our task becomes predicting the target's first token.

[4] We assume that contexts are naturally-occurring and not adversarial.